IALM - Experiment No-2 A Comparative Analysis of Regression Techniques Across Multiple Datasets

(Posted on <13 October, 2025>, Submission Deadline <19 October, 2026>)

Problem Statement

Select suitable regression techniques and apply them to multiple datasets. Train the models and compare their performance using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. Explore and visualize the relationship between variables, regression fit, actual versus predicted values, and residual errors. Use suitable graphs such as scatter plots, regression plots, residual plots, and bar charts to compare model performance across datasets. Identify the most suitable regression technique for each dataset based on the evaluation metrics and visual observations, and present your findings with appropriate graphs and conclusions.

Rubric for Grading Submission

You need to submit your report for evaluation on a scale of 10. The report shall consist of your understanding of the problem, code, results, novelty (if any) and conclusions. Following rubric shall be used for grading your submission

Marks Bracket	Details	Requirements
0	No submission within deadline	
1-4	Used code provided as it is without any changes. No novelty. Marks based on the presentation and time of submission.	Successfully installed and executed the code.
5-8	Minor changes in the code, time of submission.	Performing different data cleaning and exploration methods.
9-10	Major changesin the code, out rightly new concept, time of submission.	Trying above on any other dataset.